

COURTNEY CRAWFORD

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CAREER HISTORY

University of Sydney School of Physics Postdoctoral Researcher (Promoted to Level B in 2025)	July 2022 - present
Louisiana State University Physics & Astronomy Ph.D. in Physics	Graduation: May 2022
University of Oklahoma School of Physics & Astronomy B.S. in Astrophysics	Graduation: May 2018

HIGHLIGHTS

- 28 refereed publications, including 8 first-authored; h-index of 10
- Awarded promotion to Postdoctoral Researcher Level B
- Lead-supervisor for one PhD student, Co-supervisor for two additional PhD students
- Primary supervisor for 11 of 20 undergraduate students and two of three Honours students
- Developed lecture material for four different courses at USyd
- Lead organiser for Sydney Spectroscopy Workshop to be held in October 2026

RESEARCH SUMMARY

My research focuses on two interconnected areas: the rare **Hydrogen-deficient Carbon (HdC) stars**, which are believed to form from the mergers of two white dwarf stars, and **asteroseismology**, which uses stellar brightness variations to probe the deep interiors of stars. My current work brings these two areas together, alongside my broader expertise in time-domain astronomy, stellar evolution modelling, and spectroscopy. I have **published 8 first-authored and 20 co-authored articles** in reputable journals and hold an h-index of 10. A full list of publications is attached at the end of this document.

TEACHING & SUPERVISION

My teaching and supervision activities focus on student research and mentorship. I have supervised or co-supervised **four PhD students**, including my current role as **lead supervisor for one**, and have served as **primary supervisor for 11 undergraduate and two Honours students** from a total of 19 undergraduate and three Honours students supervised. I also contribute to course design and delivery within the USyd School of Physics.

Courses Taught/Developed

- **OLES1603: Astronomy: Exploring the Universe** *(Sem 1 2026, USyd)*
Contributed to curriculum planning and developed outline slides for the stars and galaxies lectures prior to the course's first delivery.
- **OLET1638: Astronomy from Stars to Black Holes** *(Sem 1 2025 April Intensive, USyd)*
Sole lecturer for this four-session intensive course. Substantially revised provided materials to align with updated assessments and wrote the final exam.
- **OLET1636: Astronomy from Earth to Exoplanets** *(Sem 1 2025 March Intensive, USyd)*
Sole lecturer for this four-session intensive course. Substantially revised provided materials to align with updated assessments and wrote the final exam.
- **PHYS2014/PHYS2914: Data Science in Astronomy** *(2 Lectures, USyd, 2024 Sem 2)*
Contributed to curriculum development for this new data science course, and designed and delivered two lectures and associated labs on clustering techniques applied to stellar variability.
- **ASTR 1109: Stellar Astronomy Lab** *(4 semesters, LSU)*
Sole instructor for four semesters. Developed four new labs, managed all curriculum and assessment, and received strong student evaluations.

Current Students

- **Cameron Davidson** *(PhD Student, lead-supervisor, 2026-present)*
Infrared dust measurements of R Coronae Borealis Stars
- **Lea Schimak** *(PhD Student, co-supervisor, 2023-present)*
Asteroseismology of binary red giants
- **Yingxiang Wang** *(PhD Student, co-supervisor, 2023-present)*
Simple and accurate measurements of red giant asteroseismic parameters
- **Lauren Barkey** *(Independent research, lead-supervisor, May 2026-present)*
Spectrophotometric Variability of HdC Stars

Past Students

- **Maya Carlyon** *(SSP, co-supervisor, 2026 Sem 1)*
Contamination from nearby sources in TESS photometric data
- **Julius Hamprecht & Michael Shen** *(SSP, co-supervisor with Lea Schimak, 2025 Sem 2)*
Measuring Masses of Red Giant Stars with Asteroseismology and Machine Learning
- **Michael El-Hurr** *(SSP, sole-supervisor, 2025 Sem 2)*
Temperature Class and Carbon Molecular Band Strengths of Hydrogen-deficient Carbon Stars During Pulsation
- **Jonas Káral** *(SSP, sole-supervisor, 2025 Sem 2)*
Post-merger Simulations of DY Persei Variable Stars
- **Sreenivas K.R.** *(PhD, co-supervisor Jan 2023-Mar 2025)*
Ensemble Asteroseismology with Kepler and TESS
- **Xander Buckingham** *(SSP, co-supervisor with Lea Schimak, 2025 Sem 1)*
Convective Boundaries in He-burning Stellar Models
- **Josh Ferguson** *(Dalyell, sole-supervisor, 2025 Sem 1)*
Characterisation of Radio Emitting Eclipsing Binaries
- **Isabella Treloar & Ozan Kocatepe** *(SSP, sole-supervisor, 2024 Sem 2)*
Spectroscopy of pulsations in RCB and dLHdC stars
- **Haonan Li** *(Honours, sole-supervisor, 2024)*
Anomalous peaks and potential tertiary systems found via Kepler oscillating star data
- **Jasmyn Curry** *(SSP, sole-supervisor, 2024 Sem 1)*
Spectroscopy of pulsations in RCB stars
- **Josh Ferguson** *(SSP, sole-supervisor, 2024 Sem 1)*
Dipole mode suppression in the Kepler red giant sample
- **George Feng** *(SSP, co-supervisor, 2024 Sem 1)*
A red giant in the Pleiades cluster
- **Nikita Nikultsev** *(Honours, sole-supervisor, 2023 Sem 1 & 2)*
Merging With MESA: Modelling DY Pers by Fusing White Dwarfs
- **Maxwell Bennett** *(Honours, co-supervisor, 2023 Sem 1 & 2)*
Photometry in Crowded Stellar Fields with NASA's TESS Mission
- **Noah Smith & Nayan Gallego Sivaraman** *(SSP, co-supervisor, 2023 Sem 2)*
Analysis of the emission from Low Mass X-ray Binary A0620-00 over 6 years
- **Caleb Clark** *(SSP, sole-supervisor, 2023 Sem 1)*
An automated pipeline to find RCB dust declines
- **Amelie Read** *(Denison Scholar, co-supervisor, 2023 Summer)*
Searching the Gaia and TESS data for delta Scuti stars
- **Yifan Chen** *(Dalyell, co-supervisor, 2022 Sem 2)*
Automated detection and refinement of asteroseismic parameters in 16,000 stars
- **Alexander Jephtha** *(SSP, co-supervisor, 2022 Sem 2)*
A search for oscillating wide binaries
- **Cameron Davidson** *(SSP, co-supervisor, 2022 Sem 2)*
Verifying the low-mass Kepler red giants

CONTRIBUTED & INVITED TALKS

Over 200 Years of Studying R Coronae Borealis Stars

Surveys and Photometry of Southern Stars

Invited, July 2026

Learning about low-mass DWD mergers from their post-merger remnants Seminar at Macquarie	<i>Invited, 2026</i>
The Variability of Hydrogen-deficient Giants Stars in Newcastle	2025
The Extreme Helium Stars - A New Type of Stochastic Low-Frequency Variable TASC9/KASC16	2025
Hydrogen-deficient Carbon Stars: What we know (and what we don't) Seminar at USyd	2025
Kepler Secondary Clump Stars: The Amplitude-Activity Relation Stars in Brisbane	2024
The HdC stars– updates on dust properties and the progenitor population Seminar at Armagh	<i>Invited, 2024</i>
The Dust Properties of R Coronae Borealis Stars Advancements for Cool Evolved Stars	2024
Hydrogen-deficient Carbon Stars: Advances in Formation and Dust Production Seminar at Monash	<i>Invited, 2024</i>
Hydrogen-deficient Carbon Stars– What happens after a white dwarf merger? Seminar at ANU	<i>Invited, 2023</i>
Hydrogen-deficient Carbon Stars– What happens after a white dwarf merger? Seminar at UNSW	<i>Invited, 2023</i>
Hydrogen-deficient Carbon Stars– The State of the Field Seminar at Macquarie	<i>Invited, 2022</i>
An MK-like Optical Spectral Classification Scheme for Hydrogen Deficient Carbon Stars and R Coronae Borealis Variables AAS #240	2021
3D Mapping The Distribution of R Coronae Borealis Stars AAS #238, id. 323.03	2020
Modeling Low Metallicity R Coronae Borealis Stars with MESA Understanding Dust 30 Years After CCM	<i>Invited, 2019</i>

SERVICE AND LEADERSHIP

Sydney Spectroscopy Workshop <i>Lead Organiser</i>	October 2026
Asteroseismology Group Journal Club <i>Founder and Organiser</i>	2023–present
SSP Student Telescope Visit <i>Committee Member, Trip Supervisor</i>	September 2025
Leading Teams Workshop <i>Participant, Nominated by School of Physics</i>	2025
MESA School Leuven <i>Teaching Assistant</i>	June 2025
MESA Down Under Workshop <i>Teaching Assistant, Local organising Committee</i>	June 2024
LGBTQ Community Events <i>Organiser– ASA Annual Meeting 2023, MESA Down Under</i>	2023, 2024
Sydney Institute for Astronomy (SfA) Seminar Organisation <i>Co-organiser</i>	Jan 2023–June 2024

9th Australian Exoplanet Workshop	2023
<i>Local Organising Committee Member</i>	
Physics & Astronomy Graduate Student Organisation	2019–2021
<i>President</i>	2020–2021 US Academic Year
<i>Vice President</i>	2019–2020 US Academic Year
Astronomy on Tap, Baton Rouge	2018–2022
<i>Emcee, Lead Organiser</i>	2020–2022
<i>Staff Member</i>	2018–2020
Girl’s Day at the Museum	2018–2019
<i>Physics Activity Organiser, Table Volunteer</i>	2019
<i>Table Volunteer</i>	2018

GRANTS & AWARDS

I am a **co-investigator on an ARC DP**, and was a **co-investigator on 12 telescope proposals and four computing proposals** for a total award of 28 nights of observing time (valued at roughly \$235K) and over two million KSU of computing time (valued at roughly \$36K).

Grants

- **ARC DP250102562** 2025-2027
Decoding Stellar Physics with NASA’s James Webb Space Telescope
Appointed as Co-Investigator to replace Dan Huber
- **Hunstead Workshop Award** 2026
University of Sydney School of Physics Award for ECR-led Workshop
To be held in October 2026

Computing Time

- National Computational Merit Allocation Scheme 2025
- National Computational Merit Allocation Scheme 2024
- SIH HPC Allocation Scheme 2024
- SIH HPC Allocation Scheme 2023

Telescope Observing Time

- Keck/NIRC-2 2026
- ANU 2.3M/WiFeS 2026
- ANU 2.3M/WiFeS 2025
- AAT/Veloce 2024
- ANU 2.3M/WiFeS 2024
- Keck/HIRES 2023
- ANU 2.3M/WiFeS 2023
- AAT/Veloce Project 1 *Principal Investigator, 2022*
- AAT/Veloce Project 2 2022
- ANU 2.3M/WiFeS 2022
- CTIO/KOSMOS 2019
- Gemini/NIRI *Principal Investigator, 2017*

REFERENCES

Prof. C. Simon Jeffery	simon.jeffery@armagh.ac.uk
<i>Professor</i>	
Armagh Observatory and Planetarium, College Hill, Armagh, UK	
Prof. Geoffrey Clayton	gclayton@phys.lsu.edu
<i>Ball Family Distinguished Professor</i>	

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Louisiana State University

Prof. Timothy Bedding
Professor

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Sydney Institute for Astronomy (SIfA), School of Physics,
University of Sydney, NSW 2006, Australia

8 First-Authored Journal Papers

Crawford, C. L., Jeffery, C. S., Pedersen, M. G., Bedding, T. R., Montet, B. T., Clayton, G. C. & Tisserand, P. **TESS Observations of Stochastic Low-frequency Variability in Extreme Helium Stars.** *arXiv e-prints*, arXiv:2605.23209. doi:10.48550/arXiv.2605.23209. arXiv: 2605.23209 [astro-ph.SR] (May 2026).

Crawford, C. L., Li, Y., Huber, D., Yu, J., Bedding, T. R., Martell, S. L., Montet, B. T., Stello, D., Isaacson, H., Howard, A. W., Fulton, B. J., Zhang, J., Polanski, A. S. & Weiss, L. M. **The highest mass Kepler red giants – II. Spectroscopic parameters, the amplitude–activity relation, and unexpected halo orbits.** MNRAS **542**, 3289–3301. doi:10.1093/mnras/staf1421. arXiv: 2508.12585 [astro-ph.SR] (Oct. 2025).

Crawford, C. L., Soon, J., Clayton, G. C., Tisserand, P., Bedding, T. R., Clark, C. J. & Lee, C.-U. **A comprehensive study of the dust declines in R Coronae Borealis stars.** MNRAS **537**, 2635–2646. doi:10.1093/mnras/staf215. arXiv: 2412.16393 [astro-ph.SR] (Mar. 2025).

Crawford, C. L., Nikultsev, N., Clayton, G. C., Tisserand, P., Soon, J. & Pedersen, M. G. **Modelling hydrogen-deficient carbon stars in MESA - the effects of total mass and mass ratio.** MNRAS **534**, 1018–1027. doi:10.1093/mnras/stae2149. arXiv: 2408.09700 [astro-ph.SR] (Oct. 2024).

Crawford, C. L., Bedding, T. R., Li, Y., Stello, D., Huber, D., Yu, J., Sreenivas, K. R., Li, T. & Kerrison, E. F. **The highest mass Kepler red giants - I. Global asteroseismic parameters of 48 stars.** MNRAS **528**, 7397–7410. doi:10.1093/mnras/stae473. arXiv: 2402.07380 [astro-ph.SR] (Mar. 2024).

Crawford, C. L., Tisserand, P., Clayton, G. C., Soon, J., Bessell, M., Wood, P., García-Hernández, D. A., Ruiter, A. J. & Seitzzahl, I. R. **A spectral classification system for hydrogen-deficient carbon stars.** MNRAS **521**, 1674–1699. doi:10.1093/mnras/stad324. arXiv: 2210.04416 [astro-ph.SR] (May 2023).

Crawford, C. L., Tisserand, P., Clayton, G. C. & Munson, B. **Peculiar hydrogen-deficient carbon stars: strontium-rich stars and the s-process.** A&A **667**, A85. doi:10.1051/0004-6361/202142882. arXiv: 2112.07689 [astro-ph.SR] (Nov. 2022).

Crawford, C. L., Clayton, G. C., Munson, B., Chatzopoulos, E. & Frank, J. **Modelling R Coronae Borealis stars: effects of He-burning shell temperature and metallicity.** MNRAS **498**, 2912–2924. doi:10.1093/mnras/staa2526. arXiv: 2007.03076 [astro-ph.SR] (Oct. 2020).

20 Co-Authored Journal Papers

Wittenmyer, R. A., Venner, A., Fairnington, T., Zhou, G., Wright, D. J., Curtin, E., Bedding, T. R., **Crawford, C. L.**, Li, Y., Stello, D., Hon, M., Huber, D., Grundahl, F., Fredslund, M. S., Palte, P. L., Gan, T., Horner, J., Kielkopf, J., Kane, S. R., Plavchan, P., Shporer, A., Tinney, C. G., Zhang, H., Mengel, M. W. & Okumura, J. **The Pan-Pacific Planet Search - IX. A menagerie of companions orbiting evolved stars.** MNRAS. doi:10.1093/mnras/stag1038 (June 2026).

Chowhan, A., Bedding, T. R., Huber, D., Joel Ong, J. M., Schimak, L. S., Li, Y., **Crawford, C. L.** & White, T. R. **CHARA interferometry and TESS asteroseismology of the core-helium burning red giant κ Cyg.** MNRAS **548**, stag719. doi:10.1093/mnras/stag719. arXiv: 2604.13501 [astro-ph.SR] (May 2026).

Li, T., Bi, S., Li, Y., Wu, Y., Zhang, R., Nielsen, M., **Crawford, C. L.** & Li, G. **Ensemble Asteroseismology for 138 Kepler Metal-poor Red Giants: Stellar Fundamental Parameters.** ApJ **1003**, 49. doi:10.3847/1538-4357/ae5db9 (May 2026).

- Wang, Y., Bedding, T. R., Li, Y., Chen, Y., **Crawford, C. L.**, Huber, D. & Sreenivas, K. R. **Asteroseismic diagnostics for red giants with Kepler: measuring ϵ and small frequency separations in 16 000 stars.** MNRAS **548**, stag643. doi:10.1093/mnras/stag643. arXiv: 2604.01847 [astro-ph.SR] (May 2026).
- Schimak, L. S., Bedding, T. R., **Crawford, C. L.**, Beck, P. G., Li, Y., Huber, D., Ong, J., Montet, B. T., Pedersen, M. G., Grossmann, D. H., Mathur, S. & García, R. A. **Testing red clump models with the asteroseismic binary KIC 10841730.** MNRAS **546**, stag151. doi:10.1093/mnras/stag151. arXiv: 2601.12773 [astro-ph.SR] (Mar. 2026).
- Iles, E. J., Bland-Hawthorn, J., **Crawford, C.**, Croom, S. M., Davis, H., Pedersen, M. G., Green, A., Gunawardhana, M., Icaza-Lizaola, M., Johnston, H., Kerrison, E. F., Mai, Y., Montet, B. T., Rose, K., Rutherford, T., Saraf, M., Sirks, E. L., Spalding, E., Tuntipong, S., van de Sande, J. & Yamsiri, P. **findAbar: How astronomers may perceive the bar in galaxies differently.** PASA **42**, e166. doi:10.1017/pasa.2025.10126. arXiv: 2511.09908 [astro-ph.GA] (Nov. 2025).
- Kjeldsen, H., Bedding, T. R., Li, Y., Grundahl, F., Andersen, M. F., Wright, D. J., Soutter, J., Wittenmyer, R., Reyes, C., Stello, D., **Crawford, C.**, Zhou, Y., Clerte, M., Pallé, P. L., Simon-Diaz, S., Christensen-Dalsgaard, J., Handberg, R., Hansen, H., Heeren, P., Jessen-Hansen, J., Lund, M. N., Lundkvist, M. S., Brogaard, K., Tronsgaard, R., Rudrasingam, J., Casagrande, L., Horner, J., Huber, D., Lattanzio, J., Martell, S. L. & Murphy, S. J. **Asteroseismology of the G8 subgiant β Aquilae with SONG-Tenerife, SONG-Australia and TESS.** A&A **700**, A39. doi:10.1051/0004-6361/202554633. arXiv: 2506.00493 [astro-ph.SR] (Aug. 2025).
- Mehla, A., Kasliwal, M. M., Karambelkar, V., Tisserand, P., **Crawford, C.**, Clayton, G., Soon, J. & Bhalariao, V. **Oxygen Isotope Ratios in Hydrogen-deficient Carbon Stars: A Correlation with Effective Temperature and Implications for White Dwarf Merger Outcomes.** PASP **137**, 044201. doi:10.1088/1538-3873/adc0bf. arXiv: 2412.03664 [astro-ph.SR] (Apr. 2025).
- Sreenivas, K. R., Bedding, T. R., Huber, D., **Crawford, C. L.**, Stello, D., Pedersen, M. G., Li, Y. & Hey, D. **Testing the wavelength dependence of oscillations and granulation in red giants using Kepler and TESS.** MNRAS **537**, 3265–3275. doi:10.1093/mnras/staf220. arXiv: 2502.01899 [astro-ph.SR] (Mar. 2025).
- Li, Y., Bedding, T. R., Huber, D., Stello, D., van Saders, J., Zhou, Y., **Crawford, C. L.**, Joyce, M., Li, T., Murphy, S. J. & Sreenivas, K. R. **Realistic Uncertainties for Fundamental Properties of Asteroseismic Red Giants and the Interplay between Mixing Length, Metallicity, and numax.** ApJ **974**, 77. doi:10.3847/1538-4357/ad6c3e. arXiv: 2407.09967 [astro-ph.SR] (Oct. 2024).
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- Sreenivas, K. R., Bedding, T. R., Li, Y., Huber, D., **Crawford, C. L.**, Stello, D. & Yu, J. **A simple method to measure ν_{max} for asteroseismology: application to 16 000 oscillating Kepler red giants.** MNRAS **530**, 3477–3487. doi:10.1093/mnras/stae991. arXiv: 2401.17557 [astro-ph.SR] (May 2024).
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